

Vector Electronics

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October 30, 2025

The purpose of this article is to describe an attempt to solve electronic circuits made of non-linear devices like diodes, bipolar transistors, JFETs, MOSFETs, etc.

A common approach is to set the circuit at some operating point and wiggle around it with the input signal.

For small variations, even the most non-linear device can be approximated by a linear model. The *Small Signal Analysis*, as it is called, is a classical framework for that.

However, some circuits are trickier and make the device operate way beyond the small signal hypothesis.

The class AB (push/pull) amplifier is one example: a transistor in the pair goes from cutoff to linear region as the input changes rises. The other transistor does the opposite. Meanwhile, the output follows the input nicely. The circuit as a whole surprisingly acts as if everything were linear.

To fully analyse this, one could try to use a full model of the transistor (exponential curve) however:

- the equations usually have no closed-form expression (equations combining $\exp()$ and polynomials)
- it doesn't give much insight on *how* the thing works.

An extension of the Small Signal Analysis

A nonlinear function can always be approximated by a piecewise affine (linear) function¹, provided the model is refined and enough control points are introduced. Thus, when analyzing a circuit, one can quite reasonably assume, as a first approximation, that the characteristic of a nonlinear component is in fact linear. The parameters of the line (slope and offset) may not yet be known, but they must necessarily correspond to one of the (slope, offset) pairs among those defined in the proposed piecewise affine model.

The task then reduces to determining which pair applies at a given operating point. The existence of such a pair is "reassured" by the fact that every circuit necessarily has *at least* one operating point. It is reasonable to assume that this solution is unique, even though certain "pathological" circuits (e.g. with feedback) may admit multiple possible operating points (stable or not). We will return to this point later.

For most practical cases, the overall result is that every component can be treated as linear, thereby enabling the full use of linear analysis techniques (superposition principle, etc.) without any loss of generality.

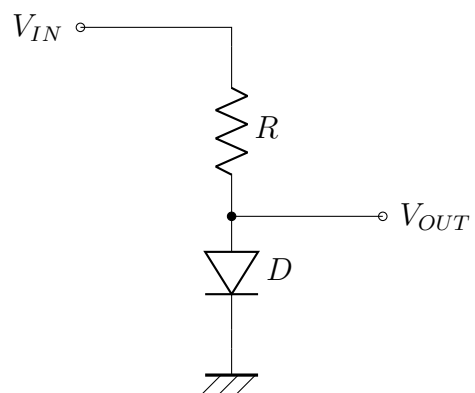
In summary, the piecewise-affine property of a nonlinear component propagates to all the properties of the electronic circuit in which it is embedded. In this sense, we speak in this paper of a *vector analysis* of the circuit. If one views a piecewise-affine function as an element of a vector space (which it is), then the circuit's properties—whatever they may be—are likewise contained within that same vector space.

It may therefore be conceivable to interpret the action of a circuit on a nonlinear component as analogous to a matrix acting on a vector. The idea that a circuit might have a kind of "matrix analogue" is perhaps worth exploring further.

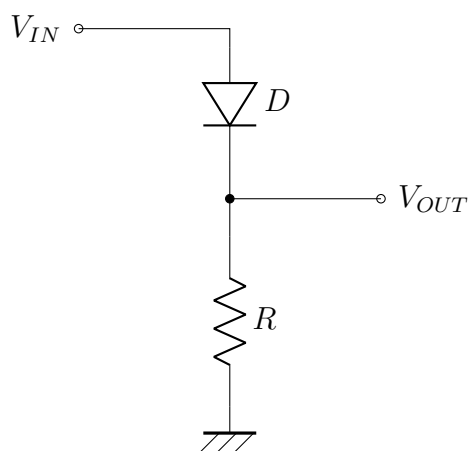
Example 1: diode and resistor I

The resistor diode circuit is the very first example of

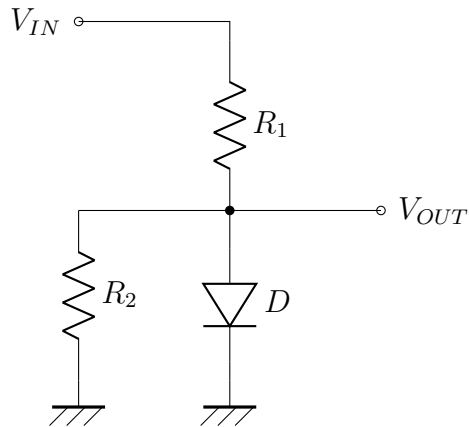
¹In other words, the set of piecewise linear functions is dense in the space of continuous real-valued functions



Example 2: diode and resistor II



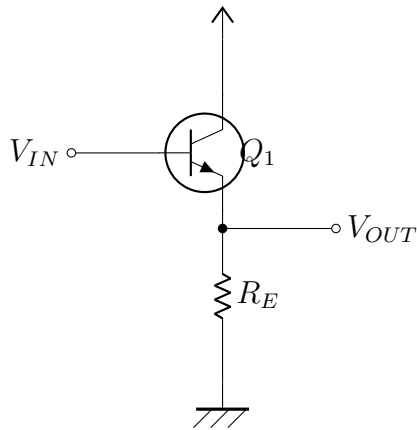
Example 3: 1 diode and 2 resistors



Example 4: common emitter

Derivation

As you will see



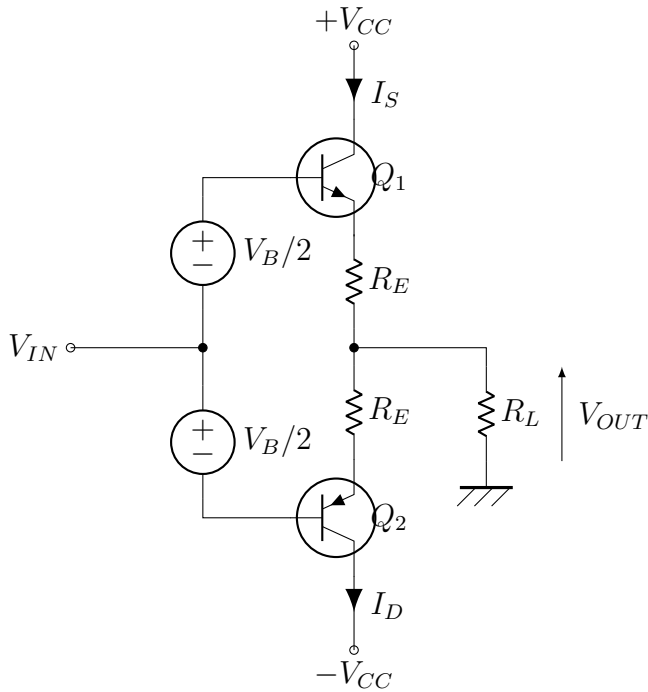
A BJT is an op amp in disguise.

We've just seen that the common emitter is basically always on. It's the very high transconductance past the $V_{BE} = 0.7V$ that makes all the system work.

Applying a very high gain to a voltage difference reminds us of op amps. Under the hood, an op amp is voltage difference amplifier with a massive gain ($> 10^6$)

The high gain is rarely used as such (except for comparators) but rather as a feedback gain. In this case, the gain doesn't matter. We just want it to be much greater than 1.

Example 5: class AB amplifier



Derivation Assuming that $I_C = f_S(V_{BE})$ for the NPN and $I_C = f_D(V_{EB})$ for the PNP we get the following:

$$\begin{cases} I_S = f_S \left(v_{IN} - v_{OUT} + \frac{v_B}{2} - R_E I_S \right) \\ I_D = f_D \left(-(v_{OUT} - v_{IN}) + \frac{v_B}{2} - R_E I_D \right) \\ V_{OUT} = R_L (I_S - I_D) \end{cases}$$

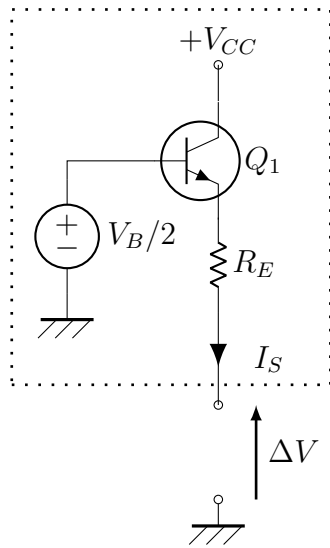
This time, the equations are trickier: the key variables I_S , I_D and V_{OUT} are all intertwined. We cannot carry out the analysis the way we did in the previous

sections.

Let's try to analyse the system as 2 independant subsystems.

Class AB subsystem: the current source

If we split the full schematic in half and set the input voltage to a constant, we can simplify the current source of the circuit to a much simpler block:



Using the usual voltage and currents laws gives the following source current:

$$I_S = f_S \left(-\Delta V + \frac{v_B}{2} - R_E I_S \right)$$

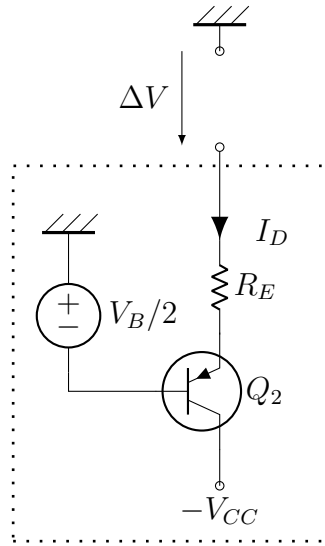
Under a linear model assumption $f_S(V) = aV + b$, we can write that:

$$I_S = a \left(-\Delta V + \frac{v_B}{2} - R_E I_S \right) + b$$

Finally, after a bit of algebra:

$$I_S = -\frac{1}{R_E} \frac{a R_E}{1 + a R_E} \Delta V + \frac{a \frac{v_B}{2} + b}{1 + a R_E}$$

Class AB subsystem: the current drain



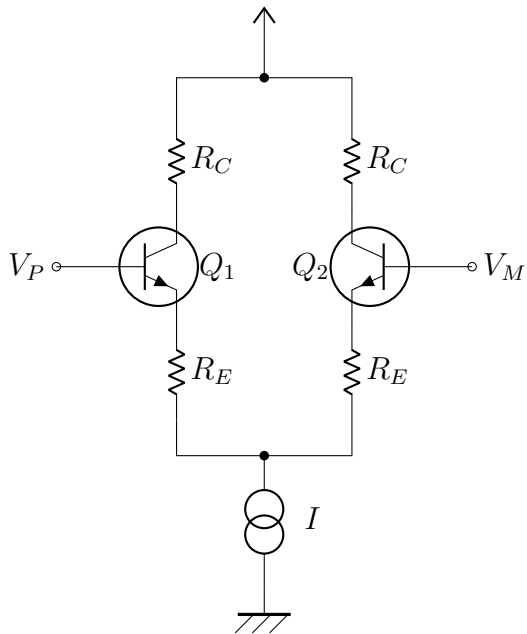
For the current drain subsystem:

Similarly, we get:

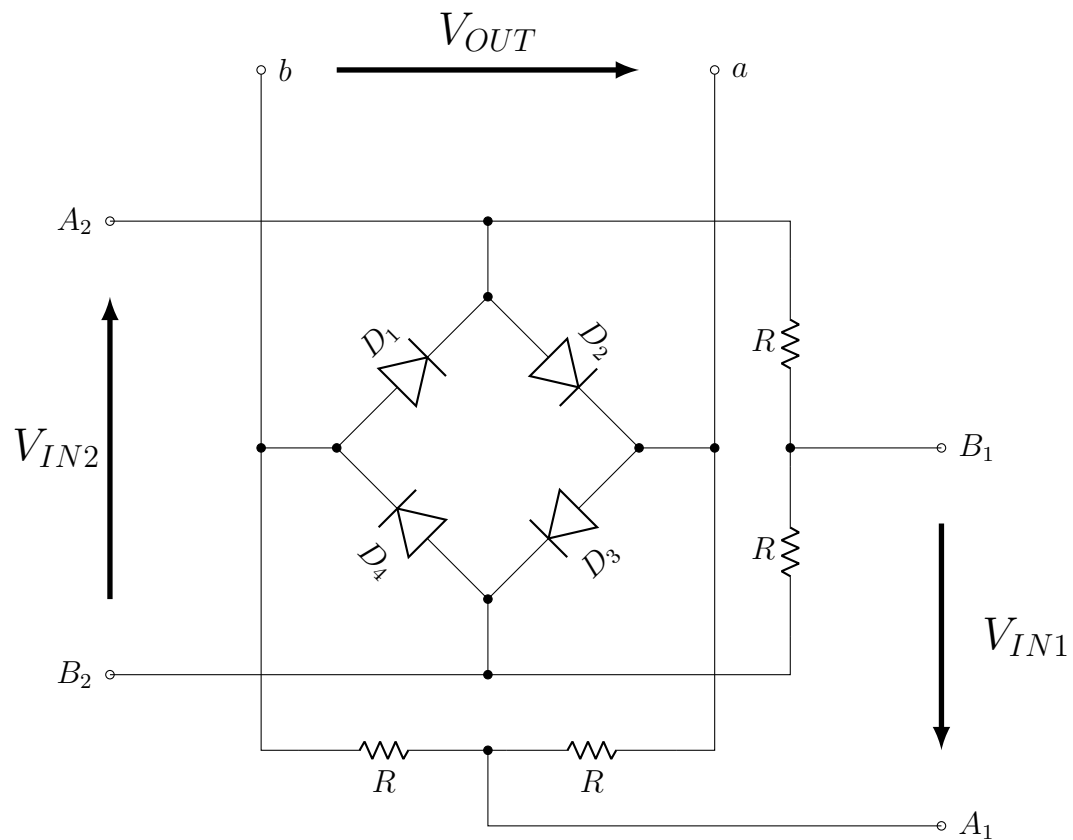
$$I_D = f_D \left(\Delta V + \frac{v_B}{2} - R_E I_D \right)$$

And:
$$I_D = \frac{1}{R_E} \frac{a R_E}{1 + a R_E} \Delta V + \frac{a \frac{V_B}{2} + b}{1 + a R_E}$$

Example 6: differential pair



Example 7: ring modulator



Example 8: the Gilbert Cell (analog multiplier)

That's a tough one, be patient...